Solution

Differentiating the function implicitly with respect to x, we get

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{a\cos t}{-a\sin t} = -\frac{\cos t}{\sin t}$$

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right)$$

$$= \frac{d}{dx}\left(\frac{-\cos t}{\sin t}\right)$$

$$= \frac{d}{dt}\left(\frac{-\cos t}{\sin t}\right)\frac{dt}{dx} = -[-\text{cosec}^2 t]\times\frac{1}{x'(t)}$$

$$= \text{cosec}^2 t \times \frac{1}{-a\sin t}$$

$$= -\frac{\cos ec^3 t}{a}\ .$$

Example 10.36

Find $\frac{d^2y}{dx^2}$ if $x^2 + y^2 = 4$.

Solution

We have $x^2 + y^2 = 4$

As before, $\frac{dy}{dx} = -\frac{x}{y}$

Hence, by the quotient rule

$$\frac{d^2y}{dx^2} = -\frac{d}{dx}\left(\frac{x}{y}\right)$$

$$= -\frac{y.1 - x.\frac{dy}{dx}}{y^2}$$

$$= -\frac{y - x\left(-\frac{x}{y}\right)}{y^2}$$

$$= -\frac{x^2 + y^2}{y^3} = -\frac{4}{y^3}\ .$$

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